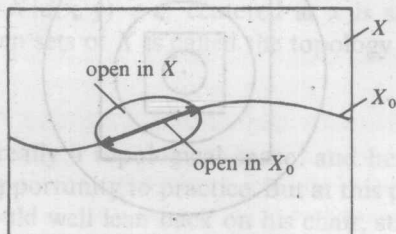


Definition (Subspace). If $(X, \mathcal{O})$ is a topological space and $X_0 \subset X$ a subset, the topology $\mathcal{O}|X_0 := \{U \cap X_0 \mid U \in \mathcal{O}\}$ on X_0 is called the *induced* or *subspace topology*, and the topological space $(X_0, \mathcal{O}|X_0)$ is called a *subspace* of $(X, \mathcal{O})$.

Instead of “open with respect to the topology of X_0 ” one says in short “open in X_0 ”, and a subset $B \subset X_0$ is then open in X_0 if and only if it is the intersection of X_0 with a set open in X :



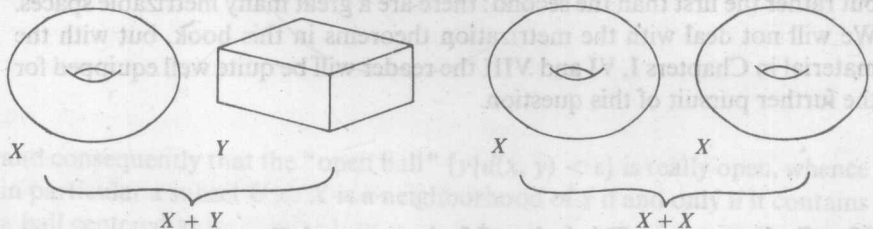
Thus such sets are not to be confused with sets “open and in X_0 ”, since they do not have to be open—open, that is, in the topology of X .

Definition (Disjoint Union of Sets). If X and Y are sets, their *disjoint union* or *sum* is defined by means of some formal trick like for instance

$$X + Y := X \times \{0\} \cup Y \times \{1\}$$

—but we immediately start treating X and Y as subsets of $X + Y$, in the obvious way.

Intuitively this operation is nothing more than the disjoint juxtaposition of a copy of X and one of Y , and we obviously cannot write this as $X \cup Y$, since X and Y do not have to be disjoint to begin with, as for example when $X = Y$ and $X \cup X = X$ consists of only one copy of X .



Definition (Disjoint Union of Topological Spaces). If $(X, \mathcal{O})$ and $(Y, \tilde{\mathcal{O}})$ are topological spaces, a new topology on $X + Y$ is given by

$$\{U + V \mid U \in \mathcal{O}, V \in \tilde{\mathcal{O}}\}$$